Some Results on Compact Sets

Definition 1 : Let (X, d) be a metric space and K be a subset of X. We say that K is a *compact subset* of X if for every open cover $\{G_\alpha : \alpha \in I\}$ of K, there are finitely many open sets $\{G_{\alpha_1}, G_{\alpha_2}, \ldots, G_{\alpha_m}\}$ with $\alpha_j \in I$ such that $K \subseteq G_{\alpha_1} \cup G_{\alpha_2} \cup \cdots \cup G_{\alpha_m}$. In other words, a subset K of metric space is compact if every open cover of the subset K has a finite subcover. If X is compact, then we say that (X, d) is a compact metric space.

Observations/Examples :

1. Every finite subset of a metric space is compact.

2. For a discrete metric space (X, d), the compact subsets are precisely finite subsets. (Solution : If K is a compact subset of a discrete metric space, then we can consider the open cover $\{\{x\} : x \in K\}$ of K. Since every open cover has a finite subcover, we see that $K \subseteq \{x_1\} \cup \{x_2\} \cup \cdots \cup \{x_m\}$. Hence, K must be finite.)

3. The closed unit interval $[0, 1]$ is a compact subset of the real line $\mathbb{R}$.

4. The open interval $(0, 1)$ is non-compact subset of the real line. (Solution : Consider the open cover $\{G_n : n \in \mathbb{N}\}$ of $(0, 1)$, where $G_n = (\frac{1}{2n}, 1)$. As $G_n \subseteq G_{n+1}$, we see that this open cover cannot have any finite subcover.)

Theorem 1 : Every compact subset is closed and bounded. (Proof: Let K be a compact subset of a metric space (X, d). We shall show that $X \setminus K$ is open. Let $a \in X \setminus K$. For each $x \in K$, let $r_x = d(x, a) > 0$. Consider the open cover $\{B(x; \frac{r_x}{2}) : x \in K\}$ of K. As K is a compact subset, there are finitely many points $x_1, \ldots, x_m \in K$ such that $K \subseteq B(x_1; r_1) \cup \cdots \cup B(x_m; r_m)$, where $r_j = \frac{r_{x_j}}{2}$. Set $r = \min\{r_1, r_2, \ldots, r_m\} > 0$.

Claim: $B(a; r) \cap K = \emptyset$. Otherwise, $y \in B(a; r) \cap K$. Thus, $y \in B(x_j; r_j)$ for some $1 \leq j \leq m$. This shows that $d(y, a) < r$ and $d(y, x_j) < r_j$. Now by triangle inequality,

$$d(a, x_j) \leq d(a, y) + d(y, x_j) < r + r_j \leq 2r_j = d(a, x_j),$$

a contradiction. Hence, $B(a; r) \subseteq X \setminus K$, showing that every point of $X \setminus K$ is an interior point. Now we shall show that K is bounded. Consider the open cover $\{B(x_0; n) : n \in \mathbb{N}\}$ of K, where $x_0 \in X$. As, $B(x_0; n) \subseteq B(x_0; n+1)$ and this open cover has a finite subcover, there exists $n_0 \in \mathbb{N}$ such that $K \subseteq B(x_0; n_0)$. Hence, K is a bounded subset.)

Theorem 2 : A closed subset of a compact metric space is compact. (Proof: Let K be a closed subset of a compact metric space (X, d). Consider an open cover $\{G_\alpha : \alpha \in I\}$ of the subset K. Then we see that $\{G_\alpha : \alpha \in I\} \cup \{X \setminus K\}$ is an open cover of X. As X is compact, there are finitely many indices $\alpha_1, \ldots, \alpha_m$ such that $X = G_{\alpha_1} \cup \cdots \cup G_{\alpha_m} \cup (X \setminus K)$. Clearly, $K \subseteq G_{\alpha_1} \cup \cdots \cup G_{\alpha_m}$. Hence, K is compact.)

Theorem 3 : Let $f\colon (X,d)\to(Y,\rho)$ be a continuous map and K be a compact subset of X. Then the image $f(K)$ is compact subset of Y. In other words, continuous image of a compact set is compact. (Proof: Consider an open cover $\{H_\alpha : \alpha\in I\}$ of the image set $f(K)$ in the metric space (Y,ρ). Since $f\colon X\to Y$ is continuous, the inverse image $f^{-1}(H_\alpha)$ of the open set H_α in Y is an open set in X. Thus we see that $\{f^{-1}(H_\alpha) : \alpha\in I\}$ is an open cover of K. As K is compact subset of X, this open cover has a finite subcover. In particular, there are finitely many indices $\alpha_1,\alpha_2,\ldots,\alpha_m\in I$ such that $K\subseteq f^{-1}(H_{\alpha_1})\cup\cdots\cup f^{-1}(H_{\alpha_m})$. Clearly, $f(K)\subseteq H_{\alpha_1}\cup\cdots\cup H_{\alpha_m}$. Hence, $f(K)$ is a compact subset of Y.)

Observations / Examples :

1. Since closed and bounded interval $[a,b]$ (for $a<b$) is homeomorphic to the unit closed interval $[0,1]$, we see that $[a,b]$ is also compact subset of the real line.

2. Let A be a closed and bounded subset of the real line $\mathbb{R}$. We shall show that A is compact. Since A is bounded, $A\subseteq[-R,R]$ for some real number $R>0$. As $[-R,R]$ is compact and A being closed subset of $\mathbb{R}$, it is a closed subset of a compact subset $[-R,R]$. Hence, by Theorem 2, A is compact. (This is known as *Heine-Borel theorem* for the real-line) .

3. A closed and bounded subset of a metric space need not be compact. Let $X=\ell^\infty$ be the space of bounded sequences with sup metric. Let $e_n\in\ell^\infty$ be given by $e_n(j)=\delta_{nj}$ for $j\in\mathbb{N}$, where the Kronecker delta $\delta_{nj}=0$ for $n\neq j$ and $\delta_{nn}=1$. Then the subset $A=\{e_n : n\in\mathbb{N}\}$ is closed and bounded subset of X. Since $||e_n-e_m||_\infty=1$ for $n\neq m$, the subset A is non-compact.

Definition 2 : A subset K of (X,d) is called *sequentially compact* if every sequence in K has a subsequence that converges in K. In other words, K is called sequentially compact if for every sequence $(x_n)_{n\in\mathbb{N}}$ in K, there is a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ such that $\lim\limits_{k\to\infty} x_{n_k}\in K$.

Theorem 4 : A compact subset is sequentially compact. (Proof: Let K be a compact subset of a metric space (X,d). Consider a sequence $(x_n)_{n\in\mathbb{N}}$ in K and suppose $A=\{x_n : n\in\mathbb{N}\}$ is the underlying set of the sequence $(x_n)_{n\in\mathbb{N}}$. Suppose the given sequence has no subsequence convergent in K. Then every $a\in K$ is not an accumulation point of A. Thus, there is an open ball $B(a;r_a)$ such that $B(a;r_a)\cap(A\setminus\{a\})=\emptyset$. We see that $\{B(a;r_a) : a\in K\}$ is an open cover of K and as K is compact, there are finitely many points $a_1,a_2,\ldots,a_m\in K$ such that $K\subseteq B(a_1,r_1)\cup B(a_2,r_2)\cup\cdots\cup B(a_m,r_m)$, where $r_j=r_{a_j}$. Clearly, there is a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ of the given sequence (x_n) such that $x_{n_k}\in B(a_j,r_j)$ for some fixed j. This shows that $x_{n_k}=a_j$ for all $k\in\mathbb{N}$. Now the subsequence (x_{n_k}) being constant is convergent in K, which is a contradiction to the above assumption. Hence, every sequence in K must have a subsequence convergent in K.)

Definition 3 : Let A be a subset of a metric space (X,d) and let $\mathcal{U}=\{G_\alpha : \alpha\in I\}$ be an open cover of A. A real number $\lambda>0$ is called a *Lebesgue number* of the covering

$\mathcal{U}$ if for every $a \in A$, there is $G_\beta \in \mathcal{U}$ ($\beta \in I$ depends upon the point $a \in A$) such that $B(a;\lambda) \subseteq G_\beta$.

We can see that not every open cover of a subset has a Lebesgue number. Let $A = (0,1)$ be an open interval in the real line $\mathbb{R}$ and let $\mathcal{U} = \{(\frac{1}{n+1}, 1) : n \in \mathbb{N}\}$ be an open cover of A. The open cover $\mathcal{U}$ of $A = (0,1)$ has no Lebesgue number. For if $\lambda > 0$, then we can take $a = \frac{\lambda}{2} \in A$. Now $B(a;\lambda) = (-\frac{\lambda}{2}, \frac{3\lambda}{2})$ is not contained in any open interval $(\frac{1}{n+1}, 1)$.

Theorem 5 : Every open cover of a compact set has a Lebesgue number. (Proof: Let K be a compact subset of a metric space (X,d) and $\mathcal{U} = \{G_\alpha : \alpha \in I\}$ be an open cover of K. For each $a \in K$, there is an open set $G_{\alpha(a)} \in \mathcal{U}$ ($\alpha(a)$ depends on $a \in K$) such that $a \in G_{\alpha(a)}$. Thus there is an open ball $B(a;r_a) \subseteq G_{\alpha(a)}$. Consider the open cover $\{B(a;\frac{r_a}{2}) : a \in K\}$ of K. As K being compact, there are finitely many points $a_1, a_2, \ldots, a_m \in K$ such that

$$K \subseteq \bigcup_{j=1}^{m} B(a_j; \frac{r_{a_j}}{2}).$$

Let $\lambda = \min\{\frac{r_{a_j}}{2} : 1 \leq j \leq m\} > 0$. If $b \in K$, then there is an i with $1 \leq i \leq m$ such that $b \in B(a_i; \frac{r_{a_i}}{2})$.

Claim : $B(b;\lambda) \subseteq B(a_i; r_{a_i}) \subseteq G_{\alpha(a_i)}$. If $y \in B(b;\lambda)$, then

$$d(y,a_i) \leq d(y,b) + d(b,a_i) < \lambda + \frac{r_{a_i}}{2} \leq r_{a_i},$$

showing that $y \in B(a_i; r_{a_i})$. Hence the given open cover of the compact set has a Lebesgue number.)

Theorem 6 : Every open cover of a sequentially compact set has a Lebesgue number. (Proof: Let K be as in the last theorem except K be sequentially compact instead of compact. Suppose there is an open cover $\mathcal{U} = \{G_\alpha : \alpha \in I\}$ of K that has no Lebesgue number. Thus for each $\lambda > 0$, there is $a \in K$ such that $B(a;\lambda) \nsubseteq G_\alpha$ for all $\alpha \in I$. Now taking $\lambda = \frac{1}{n}$ for $n \in \mathbb{N}$, there is an element $a_n \in K$ such that $B(a_n; \frac{1}{n}) \nsubseteq G_\alpha$ for all $\alpha \in I$. Since K is sequentially compact, the sequence (a_n) has a subsequence $(a_{n_k})_{k\in\mathbb{N}}$ such that $\lim_{k\to\infty} a_{n_k} = p \in K$. Now choose $\beta \in I$ such that $p \in G_\beta$. Since G_β is open, there is an $r > 0$ such that $B(p;r) \subseteq G_\beta$. Also, as $\lim_{k\to\infty} a_{n_k} = p \in K$, there is a $k_0 \in \mathbb{N}$ such that $a_{n_k} \in B(p; \frac{r}{2})$ for all $k \geq k_0$. Now choose $k \geq k_0$ such that $\frac{1}{n_k} < \frac{r}{2}$.

Claim : $B(a_{n_k}; \frac{1}{n_k}) \subseteq B(p;r) \subseteq G_\beta$. If $y \in B(a_{n_k}; \frac{1}{n_k})$, then

$$d(y,p) \leq d(y,a_{n_k}) + d(a_{n_k},p) < \frac{1}{n_k} + \frac{r}{2} \leq r,$$

showing that $y \in B(p;r)$. But by assumption $B(a_{n_k}; \frac{1}{n_k}) \nsubseteq G_\alpha$. This contradiction proves the theorem.) In view of Theorems 4 and 6, the Theorem 5 is redundant.

Theorem 7 : A sequentially compact subset is compact. (Proof: Let K be a sequentially compact subset of a metric space (X, d) and let $\mathcal{U} = \{G_\alpha : \alpha \in I\}$ be an open cover of K. From Theorem 6, the open cover $\mathcal{U}$ has a Lebesgue number, say $\lambda > 0$. Consider $a_1 \in K$. The open ball $B(a_1; \lambda) \subseteq G_{\alpha_1}$ for some $\alpha_1 \in I$. If $K \subseteq G_{\alpha_1}$, we obtain a finite subcover $\{G_{\alpha_1}\}$. Otherwise, there is $a_2 \in K \setminus G_{\alpha_1}$. Note that $d(a_1, a_2) \geq \lambda$. Also, the open ball $B(a_2; \lambda) \subseteq G_{\alpha_2}$ for some $\alpha_2 \in I$. If $\{G_{\alpha_1}, G_{\alpha_2}\}$ is an open cover of K, we are done. Otherwise, we have a point $a_3 \in K \setminus (G_{\alpha_1} \cup G_{\alpha_2})$. If the open cover $\mathcal{U}$ of K has no finite subcover, then continuing in the above manner, we get a sequence (a_n) in K with the property that $B(a_n; \lambda) \subseteq G_{\alpha_n}$ and $a_{n+1} \in K \setminus (G_{\alpha_1} \cup \cdots \cup G_{\alpha_n})$ for all $n \in \mathbb{N}$. Clearly, the sequence (a_n) in K has no convergent subsequence as $d(a_i, a_j) \geq \lambda$ for all $i \neq j$. This is a contradiction to the fact that K was sequentially compact. Thus the given open cover must have a finite subcover.)

The Theorems 4 and 7 show that the notion of compactness is equivalent to that of sequential compactness in a metric space.

Remark : If A and B are sequentially compact subsets of metric spaces (X, d) and (Y, ρ), respectively, then it can be shown that $A \times B$ is also sequentially compact (and hence, compact) subset of $(X \times Y, d_E)$. This shows that an n-cube

$$[a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n] \subseteq \mathbb{R}^n \qquad (a_i \leq b_i \text{ for } 1 \leq i \leq n),$$

is a (sequentially) compact subset of the Euclidean space $\mathbb{R}^n$. Moreover, it can be easily deduced that compact subset of the Euclidean space $\mathbb{R}^n$ are precisely the closed and bounded subset of $\mathbb{R}^n$. (This is known as the *Heine-Borel theorem.*)

Definition 4 : A subset A of a metric space (X, d) is said to be *totally bounded* if for every $\epsilon > 0$, there are finitely many points $x_1, x_2, \ldots, x_m \in A$ such that

$$A \subseteq B(x_1; \epsilon) \cup B(x_2; \epsilon) \cup \cdots \cup B(x_m; \epsilon).$$

Observations / Examples :

1. A totally bounded subset is bounded. (Solution : Let A be totally bounded. Then for $\epsilon = 1$, there are finitely many points $x_1, x_2, \ldots, x_m \in A$ such that $A \subseteq B(x_1; 1) \cup B(x_2; 1) \cup \cdots \cup B(x_m; 1)$. Let $r = \max\{d(x_i, x_j) : 1 \leq i, j \leq m\}$. If $y \in A$, then $y \in B(x_j; 1)$ for some j. Now

   $$d(y, x_1) \leq d(y, x_j) + d(x_j, x_1) < 1 + r,$$

   implies that $A \subseteq B(x_1; r + 1)$.)

2. A bounded subset of the Euclidean space $\mathbb{R}^n$ is totally bounded.

3. A bounded subset need not be totally bounded. (Solution : The subset $A = \{e_n : n \in \mathbb{N}\}$ of the sequence space ℓ^∞ is bounded but not totally bounded.)

Theorem 8 : A compact subset of a metric space is totally bounded. (Proof: Let K be a compact subset of a metric space (X, d). For any $\epsilon > 0$, consider the open cover $\mathcal{U} = \{B(a; \epsilon) : a \in K\}$ of K. As K is compact, there are finitely many $a_1, a_2, \dots, a_m \in K$ such that $K \subseteq B(a_1; \epsilon) \cup B(a_2; \epsilon) \cup \cdots \cup B(a_m; \epsilon)$. Hence K is totally bounded.)

Theorem 9 : (Generalized Heine-Borel Theorem) Let (X, d) be a complete metric space. A subset K of (X, d) is compact if and only if K is closed and totally bounded. (Proof: We have already seen that compact subsets are closed and totally bounded. Conversely, suppose K is a closed and totally bounded subset of a complete metric space (X, d). We shall show that K is sequentially compact. Consider a sequence $(x_n)_{n\in\mathbb{N}}$ in K. Let $\epsilon_k = \frac{1}{k}$ for $k \in \mathbb{N}$. As K is totally bounded, it is covered by finitely many open balls of radius $\epsilon_1 = 1$. Thus there is an open ball $B(p_1; \epsilon_1)$ that contains infinitely many terms of the given sequence. Choose, $x_{n_1} \in K_1 = K \cap B(p_1; \epsilon_1)$. Again, covering K with finitely many balls of radius $\epsilon_2 = \frac{1}{2}$, we see that an open ball $B(p_2; \epsilon_2)$ contains infinitely many terms of the subsequence in K_1. Choose, $x_{n_2} \in K_2 = K_1 \cap B(p_2; \epsilon_2)$ with $n_2 > n_1$. Continuing in this manner, we obtain a subsequence $(x_{n_j})_{j\in\mathbb{N}}$ such that $x_{n_j} \in K_j = K_{j-1} \cap B(p_j; \epsilon_j)$ and K_j contains infinitely many terms of the given sequence.

Claim : The subsequence $(x_{n_j})_{j\in\mathbb{N}}$ is a Cauchy sequence in X. For $\varepsilon > 0$, choose k_0 so large that $\frac{2}{k_0} < \varepsilon$. Now for $k, j \geq k_0$, we see that $d(x_{n_j}, x_{n_k}) \leq \text{diam}(B(p_{k_0}; \epsilon_{k_0})) = \frac{2}{k_0} < \varepsilon$. Since X is complete, the Cauchy sequence $(x_{n_j})_{j\in\mathbb{N}}$ is convergent to some point $p \in X$. But K is a closed set and $x_n \in K$ implies that $p \in K$. This shows that every sequence in K has a subsequence that converges in K. Hence, K is (sequentially) compact.)